

# Reservoir Seepage Quantification Using GRACE and Satellite Altimetry

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## Introduction

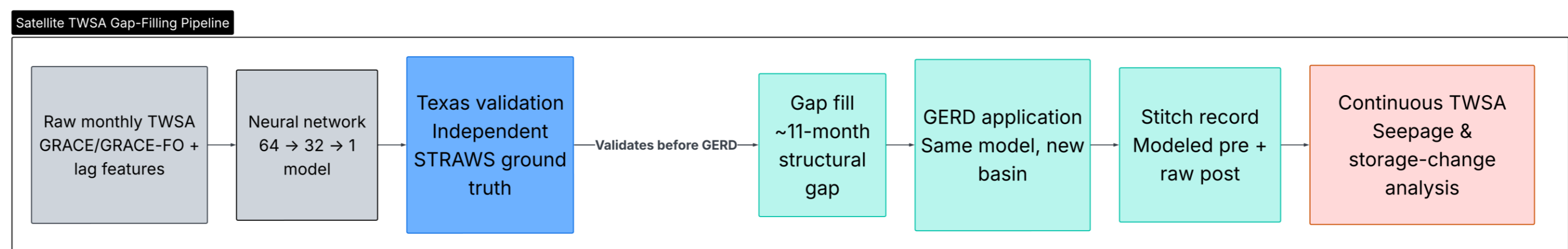
Reservoir seepage, water lost through a dam's foundation or abutments, threatens dam safety and reduces water available downstream. It's hard to measure directly, especially in transboundary basins where ground monitoring is limited or politically restricted. GRACE and GRACE-FO satellites track monthly changes in total water storage across a basin, but the record has gaps, most notably an 11-month gap between the two missions, that break continuity during critical periods like active reservoir filling. This project fills those gaps using a neural network trained on the record's trend and seasonal patterns, building on prior methods (Longuevergne et al., 2013) for isolating reservoir signals from GRACE data. Satellite altimetry works as independent water level and surface area measurements. It is used to validate and constrain the coarser GRACE mass anomalies, keeping the reconstructed record grounded in an independent check rather than relying on GRACE alone. We demonstrate this on the Grand Ethiopian Renaissance Dam (GERD), a rapidly filling reservoir central to an ongoing Nile water dispute, where a reliable, independent storage record matters for real-world monitoring.

## Research Goal

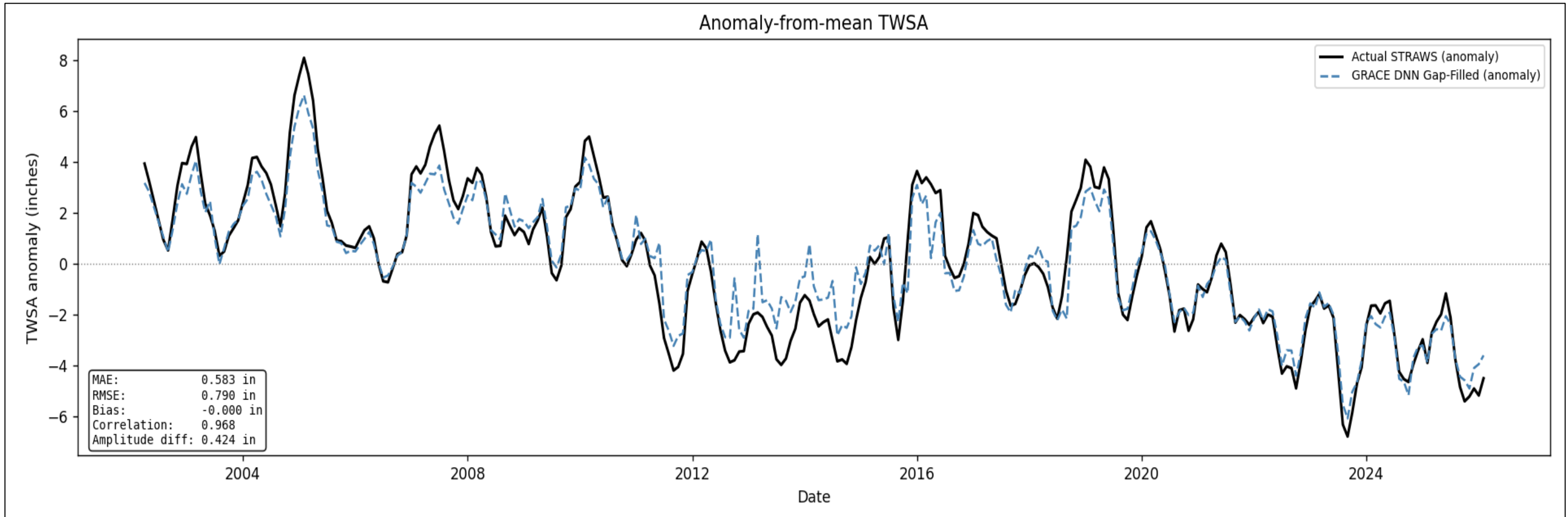
To develop and demonstrate a satellite-based framework that isolates and quantifies reservoir seepage at the Grand Ethiopian Renaissance Dam (GERD), allowing continuous and independently verifiable water storage monitoring in a transboundary basin without reliance on ground-based instrumentation.

## Methods

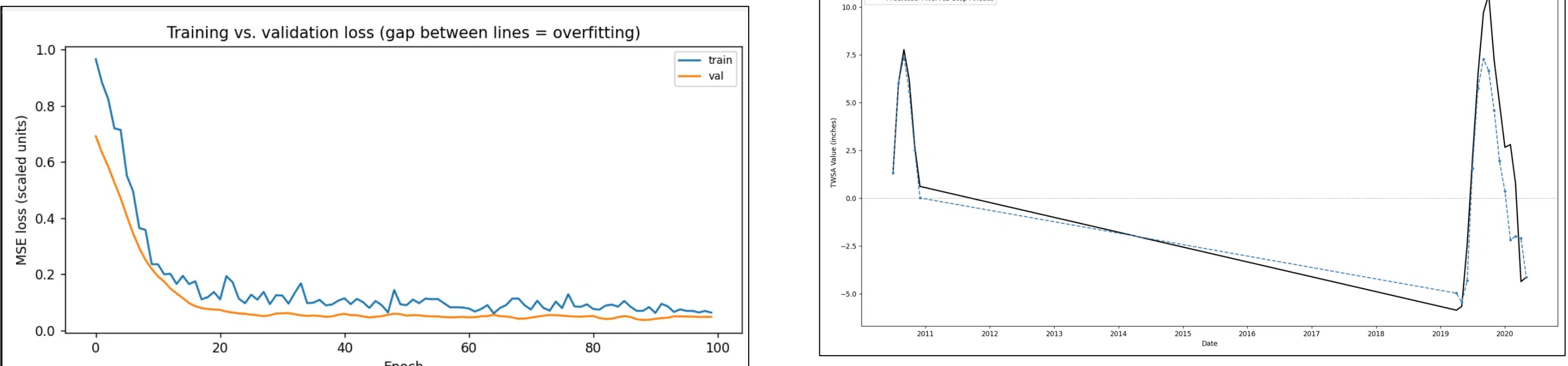
This project fills gaps in monthly GRACE/GRACE-FO terrestrial water storage anomaly (TWSA) time series using a neural network trained on lag features of the historical record. We first validated the approach against STRAWS gap-filled ground truth data over Texas, confirming the model could reliably reconstruct known values. Encouraged by this performance, we extended the same gap-filling method to the GERD basin on the Blue Nile. To avoid learning across the structural break caused by reservoir impoundment, the network was trained only on pre-impoundment data, then the real post-impoundment record (confirmed gap-free) was stitched back on to produce one continuous TWSA series spanning before and after GERD's construction. This complete record was then used for seepage and overall storage-change analysis.



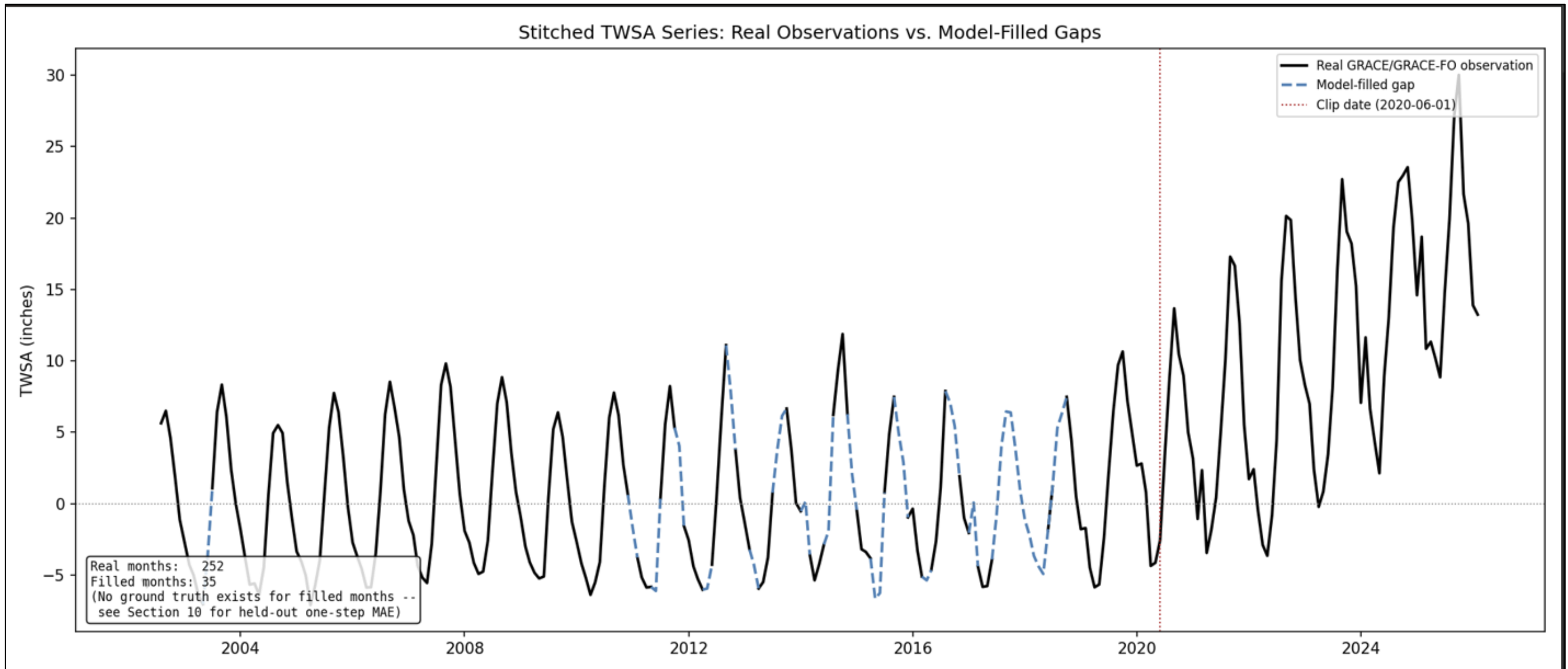
## Figures and Results



To check our approach against an established alternative, we compared the neural network's reconstructed TWSA to STRAWS, a Texas water storage record built from the same CSR/JPL GRACE data but using trend-seasonal decomposition rather than machine learning (Rateb et al. 2022). The two methods show strong agreement (MAE = 0.583 in, RMSE = 0.790 in, correlation = 0.968), supporting that the neural network behaves consistently with a standard technique before being applied to GERD, where no equivalent independent reconstruction is available to check against.

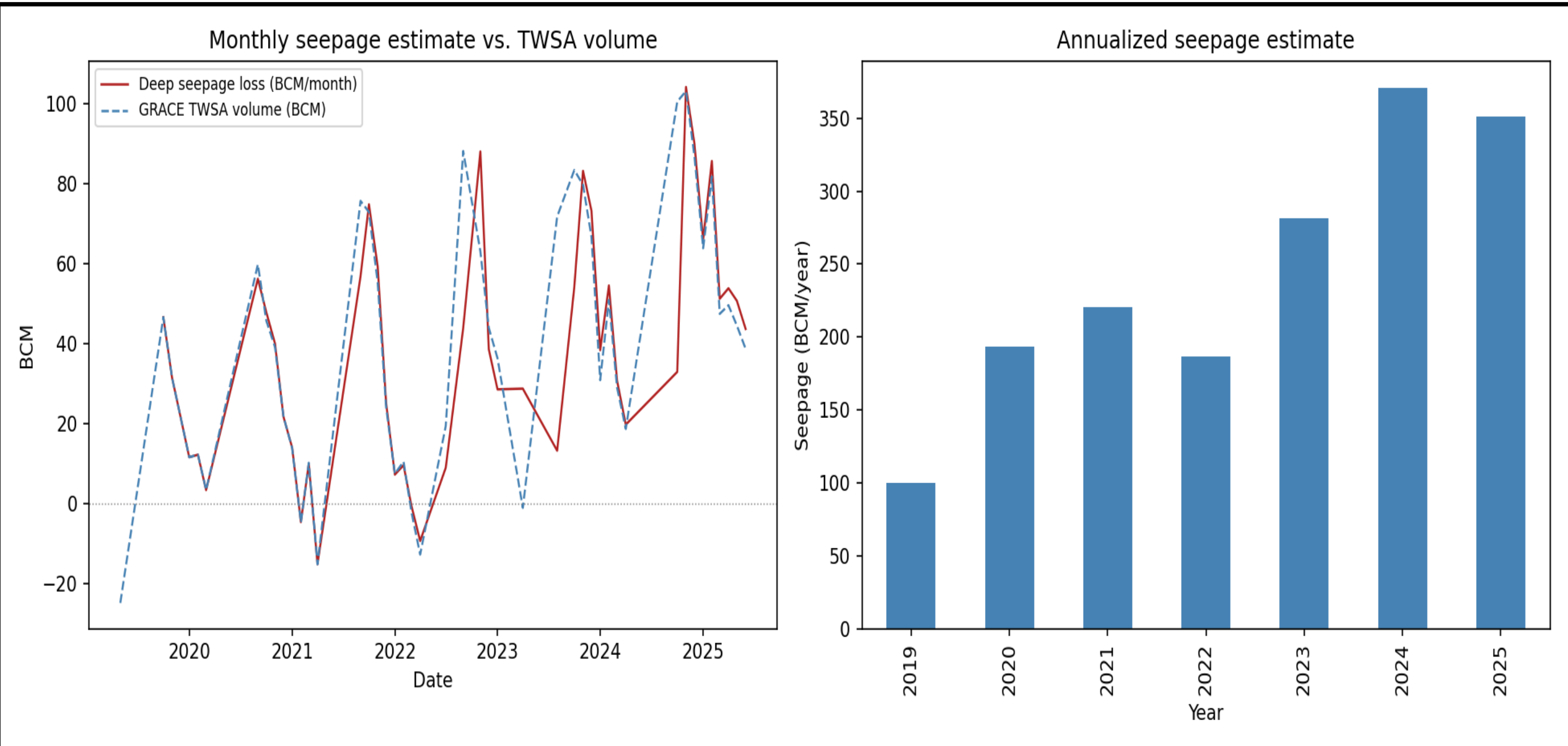


Training and validation loss converge over 100 epochs with no substantial widening gap, indicating the model is not overfitting to the GERD training data. The corresponding one-step-ahead prediction plot shows the model closely tracking actual GERD TWSA when given true historical values at each step. Notably, this measures one-step prediction error only; the recursive multi-month gap-filling used to reconstruct the ~11-month GERD data gap can accumulate larger error across a gap than a single-step forecast.



The final reconstructed TWSA record spans 2003–2025, combining real GRACE/GRACE-FO observations (black) with model-filled months (blue dashed) across the historical record's data gaps, including an 11-month gap coinciding with the period around GERD impoundment. The model was trained only on pre-impoundment data to avoid learning across this structural break; all data from July 2020 onward is raw, unmodeled GRACE-FO observation. A clear upward shift in TWSA is visible beginning around the clip date, consistent with the reservoir accumulating water mass as GERD filled. This is the physical signal the seepage analysis in the following section is built on.

## Conclusion



The neural network successfully reconstructed missing TWSA values in both the Texas validation case and the GERD record, producing a continuous, gap-free time series that clearly captures the onset of reservoir filling in 2020. This record provides an independent, satellite-based check on basin water storage that doesn't rely on ground instrumentation. Isolating a seepage-specific signal from basin-wide GRACE data proved limited by GRACE's spatial resolution relative to the reservoir's footprint — future work should explore downscaling methods or complementary local measurements to disaggregate the dam-scale signal from broader basin hydrology.

## Acknowledgments

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## References

Longuevergne et al. (2013). *GRACE water storage estimates for the Middle East and other regions with significant reservoir and lake storage*. Hydrology and Earth System Sciences, 17(12), 4817–4830.